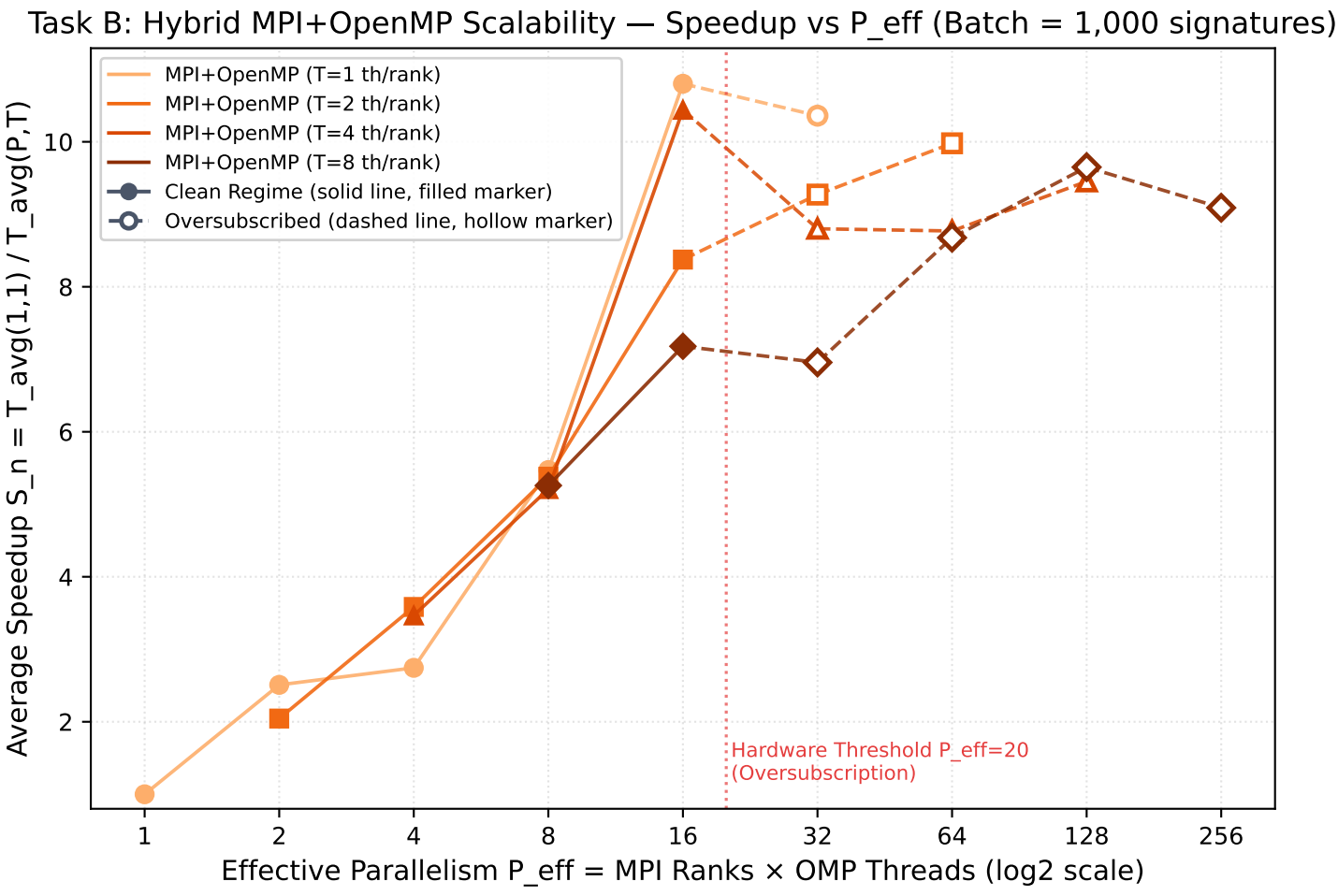
Task B: Hybrid MPI+OpenMP Scalability — Speedup vs P_eff (Batch = 1,000 signatures)
MPI+OpenMP (T=1 th/rank)
MPI+OpenMP (T=2 th/rank)
MPI+OpenMP (T=4 th/rank)
MPI+OpenMP (T=8 th/rank)
Clean Regime (solid line, filled marker)
Oversubscribed (dashed line, hollow marker)
Average Speedup S_n = T_avg(1,1) / T_avg(P,T)
2
4
6
8
10
Hardware Threshold P_eff=20
(Oversubscription)
1
2
4
8
16
32
64
128
256
Effective Parallelism P_eff = MPI Ranks × OMP Threads (log2 scale)